

Problem 12.1 (Theoretical/MATLAB)

Basics of hierarchical clustering algorithm

The matrix of distances between five multivariate observations is

$$D = \begin{bmatrix} 0 & & & & \\ 4 & 0 & & & \\ 6 & 9 & 0 & & \\ 1 & 7 & 10 & 0 & \\ 6 & 3 & 5 & 8 & 0 \end{bmatrix}$$

Using the hierarchical agglomerative clustering algorithm, cluster the observations and draw dendrograms for, respectively,

- single linkage
- complete linkage
- average linkage
- centroid linkage
- and compare the results

Problem 12.2 (Theoretical/MATLAB)

Basics of K-means clustering algorithm

The data matrix for four bivariate observations (e.g. called # 1,2,3,4) is

$$X = \begin{bmatrix} 5 & 4 \\ 1 & -2 \\ -1 & 1 \\ 3 & 1 \end{bmatrix}$$

Use the K-means algorithm to cluster the observations into K = 2 clusters for the following initial clusterings

- (12), (34)
- (13), (24)
- (1), (234)
- (123), (4)
- compare the final clusters and comment

Problem 12.3 (MATLAB)

Hierarchical and K-means clustering of university rankings

The file '***dataset_problem_12_3.mat***' contains ranking data for 25 universities in the USA, some private and some public (state) universities. The measured variables are:

- X_1 : average score in test of students just started
- X_2 : percentage of just started students in top 10% of high school class
- X_3 : percentage of applicants accepted
- X_4 : student-faculty ratio
- X_5 : estimated annual expenses
- X_6 : graduation rate

The names of the 25 universities are given in '***university_names_problem_12_3.mat***'.

It is desired to investigate whether the 25 universities naturally can be grouped into a number of clusters based on the measured variables.

Load and preprocess data:

- Load the two files
- Since the 6 variables have different units and different dynamic range, standardize the raw data matrix, $\mathbf{X}$, to a matrix, $\mathbf{Z}$, who's columns (variables) are all zero-mean and unit-variance

Hierarchical clustering (using D):

- Calculate the distance matrix, $\mathbf{D}$, for $\mathbf{Z}$
- Calculate and plot the "raw" dendrograms for, respectively,
 - o single linkage
 - o complete linkage
 - o average linkage
 - o centroid linkage
- Looking at the four dendrograms, suggest a value for K, the number of clusters, for each of the linkage methods (obviously no unique answers exist here, but it is desired to find a "reasonably" low value for K and also to avoid single universities forming their own clusters)
- Indicate which universities are in each of the clusters for each method and, if possible, comment on the groupings.

K-means clustering (using Z):

- Use the K-means algorithm to group the universities into K = 3, 4, 5 and 6 clusters, respectively.
- For each value of K, allow several different initial clusterings ("replicates"), thus allowing the algorithm to output the final clustering with lowest Within-Sum-of-Squares.
- Indicate which universities are in each of the clusters for each value of K, if possible, comment on the groupings, and compare with above.

Conclusion:

- Do you think the data indicate a natural grouping of the universities ?

Problem 12.4 (MATLAB)

Gaussian Mixture Model clustering of city crime rates data

The file '***dataset_problem_12_4.mat***' contains crime rate data for 16 cities in the USA for a specific year. The measured variables are:

- X_1 : number of murders/manslaughters pr. 100.000 inhabitants
- X_2 : number of rapes pr. 100.000 inhabitants

The names of the 16 cities can be found in the file '***city_names_problem_12_4.mat***'.

It is desired to investigate whether the 16 cities can be grouped into a few clusters with respect to crime rate data.

Load and view data:

- Load the two files
- Make a scatter plot of the observations

Gaussian Mixture Model with K = 2 components:

- Fit a GMM with $K = 2$ components to the data, that is calculate maximum likelihood estimates of the two mixing proportions, the two mean vectors and the two covariance matrices in the model. Allow several initial estimates ("replicates"), thus allowing the algorithm to output the final estimates with highest likelihood.
- Calculate and plot the posterior probabilities for each observation being member of each of the $K = 2$ clusters
- Assign the observations (cities) to $K = 2$ clusters using the MAP (Maximum A posteriori) criterion
- Interpret, if possible, the formed grouping of cities

Gaussian Mixture Model with K = 3 components:

- In the same way, fit a GMM with $K = 3$ components to the data
- Calculate and plot the posterior probabilities for each observation being member of each of the $K = 3$ clusters
- Assign the observations (cities) to $K = 3$ clusters using the MAP criterion
- Interpret, if possible, the formed grouping of cities

Comparison of the two GMM's:

- Which of the two GMM's do you prefer ?
- Calculate the Akaike Information Criterion (AIC) value for the two models, which model is best ?
- Calculate the Bayesian Information Criterion (BIC) value for the two models, which model is best ?